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Exploring the tenability of shared electric mobility alternatives: Will car users adopt eHUBs?

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ABSTRACT

Unlike single vehicle sharing schemes, eHUBs offer at least two of a wide range of shared electric-vehicles to meet users' diverse travel needs. Since there is limited knowledge about multi-modal electric vehicle sharing schemes, this study examines individual's intentions to use eHUBs with particular attention to car users. Based on survey responses from 3310 car users across six European countries including the Netherlands, Belgium, France, Germany, Ireland, and the United Kingdom, the study examines how each individuals' attitudes and socio-demographic variables affect their intentions to use different vehicles offered by eHUBs. Findings show that positive and negative attitudes toward shared mobility and their acceptance of alternate modes significantly affect intentions, while environmental beliefs have no statistically significant effect. Furthermore, age and gender emerged as significant predictors of intentions, and car users showed higher propensity to use shared electric-cars and electric-bikes over electric-cargo bikes. Based on the findings, the study presents recommendations of ways to encourage and incentivise private car users' to use shared mobility alternatives.

1. Introduction

The rapid development of economies has resulted in an increase in private vehicle ownership and use which has significant negative impacts, including environmental pollution, traffic congestion, and parking problems in major cities (Hu and Creutzig, 2022). In recent years, the concept of the shared economy has gained considerable attention, and within that, the popularity of shared mobility has grown substantially due to its increasing benefits including affordability, and convenience compared to owning a private vehicle (Standing et al., 2019). Shared mobility is made possible by advances in information and communication technologies (ICT) and innovative technologies like Global Positioning System (GPS) facilitating the rental of a shared vehicles through smartphones, and mobile phone applications (Gössling, 2018; Jie et al., 2021; Standing et al., 2019). Shared mobility is a better solution over private car use as it provides an environmentally friendly solution by reducing the number of vehicles on road and consequently decreasing traffic congestion (Mouratidis, 2022). Also, it is evident that shared mobility alternatives have the potential to motivate people to minimize their private vehicle use while capitalizing on the benefits of reduced travel costs and discouraging the need for purchasing and maintaining a private car. Moreover, shared mobility is frequently regarded as an attractive substitute for private car trips, which can help to mitigate the harmful effects of transportation on the environment, ease traffic congestion, and offer economic and equity advantages (Miskolczi et al., 2021).

In recent years, the popularity of electric vehicles has grown considerably given their ability to address the environmental challenges associated with Internal Combustion Engine Vehicles (ICEVs). Unlike ICEVs, electric vehicles have zero tail-pipe emissions, reducing the amount of carbon dioxide (CO_2) and toxic pollutants including particulate matter and nitrogen dioxide into the atmosphere, thereby contributing to NetZero and improving air quality in urban areas (Manjunath and Gross, 2017). Moreover, electric vehicles operate much more quietly than ICEVs, making a significant contribution to reducing noise pollution (Campello-Vicente et al., 2017; Tsoi et al., 2023). In the

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context of shared mobility, Curtale et al. (2021) emphasized the two-dimensional benefits of electric vehicle sharing schemes compared to ICEVs, highlighting both the shift from private car ownership to shared car and the environmental advantages of using electric vehicles over those powered by fossil fuels. Amid these benefits, cities worldwide are actively promoting electric vehicle sharing initiatives through a range of incentives, including financial support, tax benefits, infrastructure development, and regulatory measures (IEA, 2021). In response, many shared mobility operators are also transitioning to EV sharing programs by incorporating electric vehicles into their fleets (Mounce and Nelson, 2019). These vehicles are often stationed at mobility hubs which are the strategically located sites within urban areas designed to provide convenient access to shared mobility services (Claasen, 2020; Liao et al., 2024). Shared e-car services offer users with benefits similar to those using traditional fuel-driven cars whilst also reducing environmental and health impacts due to the absence of toxic tailpipe emissions at point of use and quieter vehicles. In addition, sharing vehicles saves the embedded carbon in car ownership for private use given that in the UK a car or van is parked for 97% of the time on average (Bates and Leibling, 2012). Furthermore, e-cargo bikes also could substitute shorter-distance car trips for shopping or deliveries by offering cargo carrying capacity compared to pedal cycles or e-bikes (Becker and Rudolf, 2018) whilst for other trip purposes such as commuting and leisure e-scooters and e-bikes offer potential benefits to traffic network managers reducing congestion because their small-size means they occupy less road space (Bucher et al., 2019; Pucher and Buehler, 2008). Furthermore, compared to shared pedal cycles, e-bikes potentially offer the opportunity to travel longer and faster journeys particularly in hilly areas (Dill and Rose, 2012).

Unlike conventional shared vehicle schemes that often offer only a single mode, shared electric mobility hubs (eHUBs) [Bösehans et al., 2023a, 2023b] offer at least two from a range of shared electric mobility (e-mobility) options, including e-cars, e-bikes, e-scooters, and e-cargo bikes, which has the potential to cater for a diverse set of travel requirements of users (see also Bösehans et al., 2024a, 2024b). The focus of eHUBs is to replace trips previously made by private vehicles with zero-tailpipe-emission shared electric vehicles. In some pilot regions, the eHUBs offer shared vehicles from several other shared mobility providers working in collaboration with each other. Moreover, eHUBs offer a one-way station-based approach, enabling users to return their vehicles at either the same hub or any of the other shared mobility hubs that are in collaboration (Bösehans et al., 2023b; Ferrero et al., 2018). However, the operations and implementation of eHUBs are different across different pilot regions engaged in this study.

In recent years, emerging literature has focused on examining the car users' intention towards using sharing schemes with only one vehicle offer (Haustein and Jensen, 2018; Kim et al., 2015; Mitra and Hess, 2021; Mouratidis, 2022), but little is known about car users' intentions towards using shared mobility schemes when a choice of different e-vehicles are on offer (Bösehans et al., 2023a, 2023b; Claasen, 2020). This study focuses specifically on car users due to two main reasons: car use compared to public transport causes more damage per person kilometer to the environment and to date the lack of research focusing on car users choice from a location offering multi-modal shared e-mobility. Motivated by the scarcity of available literature on multi-modal vehicle sharing schemes and the fact that substituting car trips gives most environmental benefits, this research uses the data collected in the eHUBs project and analyses the car user's preference only, and thus extends the existing findings of eHUBs which studied all users (Bösehans et al., 2023b). This study enhances previous research by i) focusing on car users, the target group for shared mobility services to reduce tailpipe emissions, ii) using larger sample size consisting of data from six pilot countries in eHUBs project, iii) exploring the direct and indirect effect of socio-demographic variables on intentions using structural equation modelling (SEM) and, iv) by taking into account car users' attitudes towards using an alternative mode for their travel. This research will generate fundamental understanding which will be crucial for policy making, generating targeted strategies and interventions to encourage sustainable transportation choices, with an aim of reducing reliance on private vehicles and meeting Net Zero (European Commission, 2019) targets in the transport domain.

This paper is structured in seven sections. Section 2 provides a review of the literature and presents the research questions. Section 3 describes the methodology which includes survey design, data collection, respondents' characteristics and data analysis method used in the present study. Section 4 gives the results of analysis performed in this study, while Section 5 presents the discussions on these findings. Section 6 presents the policy implications highlighting areas for further improvement. Finally, in Section 7 conclusions are drawn and limitations of the study articulated.

2. Literature review

This section reviews previous research relevant to this study including attitudes and influences towards using shared mobility and identifies the research gap.

2.1. Intentions and adoption of shared mobility alternatives

Over the years, several theoretical models have been used to understand the intention to adopt and the adoption of shared mobility alternatives. Prominent models include the Theory of Planned Behaviour (TPB) (Ajzen, 1985), Schwartz's norm-activation model (NAM) (Schwartz, 1977), Technology Acceptance Model (TAM) (Davis, 1993), Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003) and, Diffusion Of Innovation (DOI) (Rogers, 1995). TPB emphasises the role of attitudes (either positive or negative), subjective norms (extent to which the behaviour will be approved by society), and perceived behavioral control (either ease or difficulty in carrying out the specific behaviour) on influencing behavioral intentions. The Schwartz's NAM model highlights the importance of personal norms activated by an individual's awareness of the consequences and their acceptance of responsibility, in shaping pro-social behaviours (Schwartz, 1977). For a new innovation, TAM highlights the role of perceived usefulness and ease of use, by virtue of system design, in determining attitudes toward new technologies (Davis, 1993). UTAUT, meanwhile, emphasises the importance of social influence from others already using the technology (Venkatesh et al., 2003). DOI on the other hand, provides a theoretical model for the adoption of new innovation based on perceived characteristics such as **Relative advantage** (perceived advantage of the innovation over existing options), **Compatibility** (how well the innovation aligns with the adopter's values, experiences, and needs), **Complexity** (perceived difficulty in understanding or using the innovation), **Trialability** (opportunity to experiment with it before making a decision), and **Observability** (visible tangible results produced by the innovation) (Rogers, 1995). According to the DOI theory, the diffusion of the innovation occurs gradually, with some people adopting the innovation early (i.e. innovators, early adopters, early majority) and others only when it becomes a new norm (late majority and laggards) (Rogers, 1995).

In the context of shared mobility, the constructs of TPB are frequently identified as key predictors of intentions. For example, studies on shared e-scooters and bike sharing programs revealed significant positive impact of perceived behavioural control, subjective norms and attitudes on intentions (Eccarius and Lu, 2020; Si et al., 2020). To comprehensively understand the factors influencing intentions and the adoption of shared mobility, some studies have integrated multiple psychological models. For instance, Bachmann et al. (2018) included NAM and trust into the TPB, demonstrating a positive impact of personal norms and trust on carpooling intentions. Julagasigorn et al. (2021) proposed a theoretical framework incorporating TPB, TAM, NAM, consumer perceived value, and social capital for shared

mobility research. Bösehans et al. (2023a, 2023b) used the TPB and DOI to identify the characteristics of potential users and the factors influencing the adoption of shared electric mobility hubs and reported positive attitude towards shared mobility to be one of the strongest predictors of intentions. Burghard and Dütschke (2019) employed the DOI theory alongside constructs of attitude and social norms to analyse the acceptance of electric vehicle car sharing in Germany, identifying perceived capability and social norms as significant factors. Furthermore, Burghard and Scherrer (2022) found that perceived compatibility and attitude positively affects acceptance of both car sharing and ride pooling and perceived ease of use significantly positively affects attitude consistent with a finding of Davis (1993). Kopplin et al. (2021) applied the UTAUT2 model to the shared e-scooters adoption in Germany, revealing the significant positive role of social influence and environmental belief. Similarly, Curtale et al. (2021) observed that social influence and personal attitudes were positively associated with the intention to use shared e-cars in the Netherlands. Akbari et al. (2021) extended the TAM by integrating TPB and found perceived usefulness, perceived ease of use and subjective norms to be positively associated with the intentions to use ridesharing services in Iran. ***The research mentioned above focuses on shared mobility with respect to all users but in situations where only one shared mobility was offered and not specifically for car users. Therefore, car users as a group should be investigated more specifically with due attention to multiple electric vehicle availability in shared mobility hubs and their environmental concerns related to car use.***

2.2. Influence of socio-demographic variables on shared mobility use

Findings from previous literature shows that intentions towards shared mobility vary across socio-demographic characteristics, study regions, and over different shared mobility modes. Kim et al. (2015) found that females have a higher propensity towards electric vehicle sharing schemes compared to males, and users with higher household incomes are less likely to participate in electric vehicle sharing programs. Curtale et al. (2021) found that male respondents in the Netherlands had higher intentions towards using shared e-cars than female respondents due to females perceiving lower levels of ease in using the e-cars and feeling less influenced by social norms. Curtale et al. (2021) found that elderly respondents (age >70 years) are less likely to use shared e-cars due to higher levels of anxiety, and their expectations that using electric shared cars need more effort. In Taiwan, Eccarius and Lu (2020) found that females, compared to males, are more likely to adopt and continue to use shared e-scooters. In contrast, in Korea, Ko et al. (2021) found that males are more likely to use shared mobility alternatives than females. In Australia, Jie et al. (2021) observed that the use of shared mobility was higher among females than males, and individuals with higher annual income had a higher propensity to use shared mobility compared to individuals with lower annual income. In terms of intentions towards using shared e-scooters, Mitra and Hess (2021) reported that the use of shared e-scooters is lower among old age citizens (age >65 years) and females in Canada. Dorner and Berger (2020) found that males were more interested in shared cargo-bikes than females in Austria. Significant work by (Bösehans et al., 2023a) on electric shared mobility hubs offering a choice of vehicle reveals that young adults have higher intentions towards using shared e-bikes, e-scooters and e-cars, however the effect of gender on intentions towards using a shared vehicle was not statistically significant except for females reporting higher intentions towards using shared e-bikes. ***Overall, the findings of previous studies reveal that intentions towards shared mobility vary across socio-demographic characteristics, regions, and also for different shared mobility modes. The focus so far is again related to all users than car users, which is a limitation of previous work.***

2.3. The present study

While previous studies have predominantly focused on exploring intentions to use shared mobility, it is important to highlight that none of these studies have exclusively focused on car users, who constitute the primary target group for shared mobility services when reducing tailpipe emissions to achieve Net Zero targets (see Table 1). Furthermore, a significant portion of previous research has been conducted within the context of a single mode sharing scheme, where users were limited to only one shared vehicle option. In contrast, the present study examines car users' intentions towards using eHUBs, that offers a wide range of vehicles with a minimum of two options, including e-cars, e-bikes, e-scooters, and e-cargo bikes, with a minimum of two options which effectively cater to the travel needs of a range of car users. Moreover, this study presents the results of an extensive dataset gathered from 3310 respondents across six European countries, representing diverse geographic and cultural contexts not found in the existing literature.

In addition, this analysis considers socio-demographic variables given that they are an essential component affecting car users' intentions towards using eHUBs. Therefore, to the best of the authors' knowledge, the novelty in this study is in the examination of weather, or not car users intend to use the shared services offered at eHUBs. The study seeks to answer the following key research questions.

RQ1: What are the determinants of car users' intention towards using eHUBs?
RQ2: How do socio-demographic variables affect car users' intention towards using eHUBs?
RQ3: Do the different types of vehicles offered by eHUBs appeal to car users, and who will be using them?
RQ4: In which potential areas of interest should shared mobility providers put more effort to motivate car users to use eHUBs shared services for their travel?

The findings of this study will provide valuable insights into car users' intentions towards using shared mobility from hubs offering two or more alternative vehicles to assist service providers and policy makers to implement and develop effective strategies to increase shared electric vehicles use.

3. Methodology

The research presented in this paper used the data collected by a detailed questionnaire survey that was designed and executed in the INTERREG Northwest Europe funded, eHUBs (Smart Shared Green Mobility Hubs) project (Bösehans et al., 2023a, 2023b) and found to be suitable to apply the novel analytical approach developed to address the research questions identified. A brief description of the questionnaire will be followed by descriptive statistics setting the scene for the analytical approach used in this study.

3.1. Questionnaire survey

The questionnaire survey which aimed to gather respondents' socio-demographic characteristics, their attitudes, and intentions towards using different vehicles from eHUBs, was distributed online. The attitudinal items were derived from Diffusion of Innovation (DOI) theory (Rogers, 1995) which reflected car users' attitude towards innovation (i.e. acceptance of eHUBs in our study) and TPB theory reflecting car users' attitudes towards shared mobility, car use and the environment, subjective or social norms and perceived behavioural control in terms of using the shared vehicle from eHUBs **[**For more detail regarding questionnaire survey, refer to (Bösehans et al., 2023a, 2023b)**]**. The attitudinal questions were measured on a 7-point Likert scale (where, 1 - Strongly disagree and 7 - Strongly agree) with each of these attitudinal statements having a potential influence on car users' intentions towards

Table 1
Summary of previous studies on E-shared mobility.

Study	Sample	Country/Region	Type of Scheme	Method	Shared vehicle			
					e-cars	e-bikes	e-scooters	e-Cargobikes
Mitra and Hess (2021)	All (N = 1640)	Canada	Unimodal	Weighted logistic regression model			✓	
Eccarius and Lu (2020)	University students (N = 471)	Taiwan	Unimodal	Partial Least Square- Structural equation modelling (PLS-SEM)			✓	
Kim et al. (2015)	All (N = 533)	Seoul	Unimodal	Ordered probit model	✓			
Mouratidis (2022)	All (N = 121)	Norway	Unimodal	Binary logistic regression model			✓	
Kopplin et al. (2021)	Public transport service users (N = 749)	Germany	Unimodal	PLS-SEM			✓	
Curtale et al. (2021)	All (N = 656)	Netherlands	Unimodal	Structural equation modelling (SEM)	✓			
Ko et al. (2021)	All (N = 753)	Korea	Unimodal	Logistic regression analysis		✓	✓	
Flores and Jansson (2021)	Users and non-users of shared mobility (N = 1501)	Copenhagen and Stockholm	Unimodal	Logistic Regression model		✓	✓	
Burghard and Dütschke (2019)	All (N = 1548)	Germany	Unimodal	Regression analysis	✓			
Bösehans et al. (2023b)	All (N = 503)	Amsterdam	Multimodal	Categorical Principal Component Analysis, Cluster analysis	✓	✓	✓	✓
Bösehans et al. (2023a)	All (N = 2540)	Netherland Belgium France England Germany	Multimodal	Multiple Linear Regression model	✓	✓	✓	✓
Present study	**Car users (N = 3310)**	**Netherland Belgium France Germany Ireland U.K *[England, Scotland]***	**Multimodal**	**PLS-SEM**	✓	✓	✓	✓

using different vehicles from eHUBs. Further, intentions towards different shared mobility modes were measured on a sliding scale from Extremely unlikely (0) to Extremely likely (100). The survey included four intention-related questions, asking respondents "*How likely they would be to use each of the following from an eHUB in the future if available in their city*: *an e-bike, an e-cargo bike, an e-scooter, and an electric car*".

3.2. Sample

The questionnaire survey was distributed on-line in the pilot cities in six European countries collaborating in the eHUBs project. In total 3310 respondents (car users) completed the online questionnaire survey and were from the United Kingdom (32%), Belgium (16%), the Netherlands (15%), Germany (13%), Ireland (10%), France (9%), with 5% opting not to disclose their location. Fig. 1 describes the survey respondents' characteristics. Most of the participants were young and middle aged (i.e., 47% of the participants were in the age group of 25–44 years) with an almost equal share of male and female. In terms of educational qualification, most of the participants held an undergraduate or postgraduate degree, 41% had an annual income between £20,000-£59,999 and 81% had no prior experience of using shared mobility compared to 12% who had at least some experience.

3.3. Steps in the data analysis

Principal Component Analysis (PCA) rather than Exploratory Factor Analysis was chosen in the present study as, at the time of survey development, no specific assumptions regarding underlying constructs were made. That is, questionnaire items were created to cover a broad range of constructs informed by TPB and DOI, and which have been previously applied to measure intentions to use shared mobility (e.g., **TPB**: Eccarius and Lu, 2020; Si et al., 2020; Van Veldhoven et al., 2022; **DOI:** Bokolo, 2023; Keller et al., 2018; Therrien et al., 2014), yet previously not been applied in the context of shared electric mobility hubs. Hence, there were no prior assumptions about the interrelationships between measured variables. Furthermore, high correlations between the variables suggested that PCA was the correct method (Jolliffe, 2002). To achieve an optimal component solution, Oblique rotation (Promax) was used to derive the final set of components, as there existed substantial correlation between components (i.e., >0.32) consistent with the recommendation of (Brown, 2009). The objective of PCA is to reduce the number of explanatory variables n into a few principal components k that can replace the explanatory variable without losing too much information. The principal components can be equal to the number of variables in the dataset (i.e. $k = n$), however, the principal components having Eigenvalues greater than 1 only are considered (Bivina et al., 2019). In PCA, component loadings or factor loadings indicate a particular variable's influence on the respective component. A higher loading indicates a greater influence of the variable on the component/factor, whereas a lower value indicates that the variable is not significantly influenced.

The next step was to explore the relationship between the latent variables/constructs derived using PCA, socio-demographic variables and intentions. While there are multiple methods used in the literature to examine the relationship between the latent variables [Table 1], the present study employed Structural Equation Modelling (SEM) specifically due to its ability to analyse complex models simultaneously accommodating variables which can serve as both independent and dependent variables within the same model (Hair et al., 2017; Thakkar, 2020). Additionally, SEM offers the ability to examine direct and in-direct effects simultaneously, providing a clearer understanding of the mechanisms through which the variables influence each other (Thakkar, 2020). There are two main approaches to estimating structural equation models: covariance-based SEM (CB-SEM) and variance-based SEM (such as PLS-SEM). In CB-SEM, the model is estimated using the variance-covariance matrix. Conversely, in variance-based SEM, proxies are created initially through linear combinations of observed variables, and model parameters are then estimated using these proxies (Henseler et al., 2016). Unlike CB-SEM, which requires the data to obey multivariate normality, the partial least square

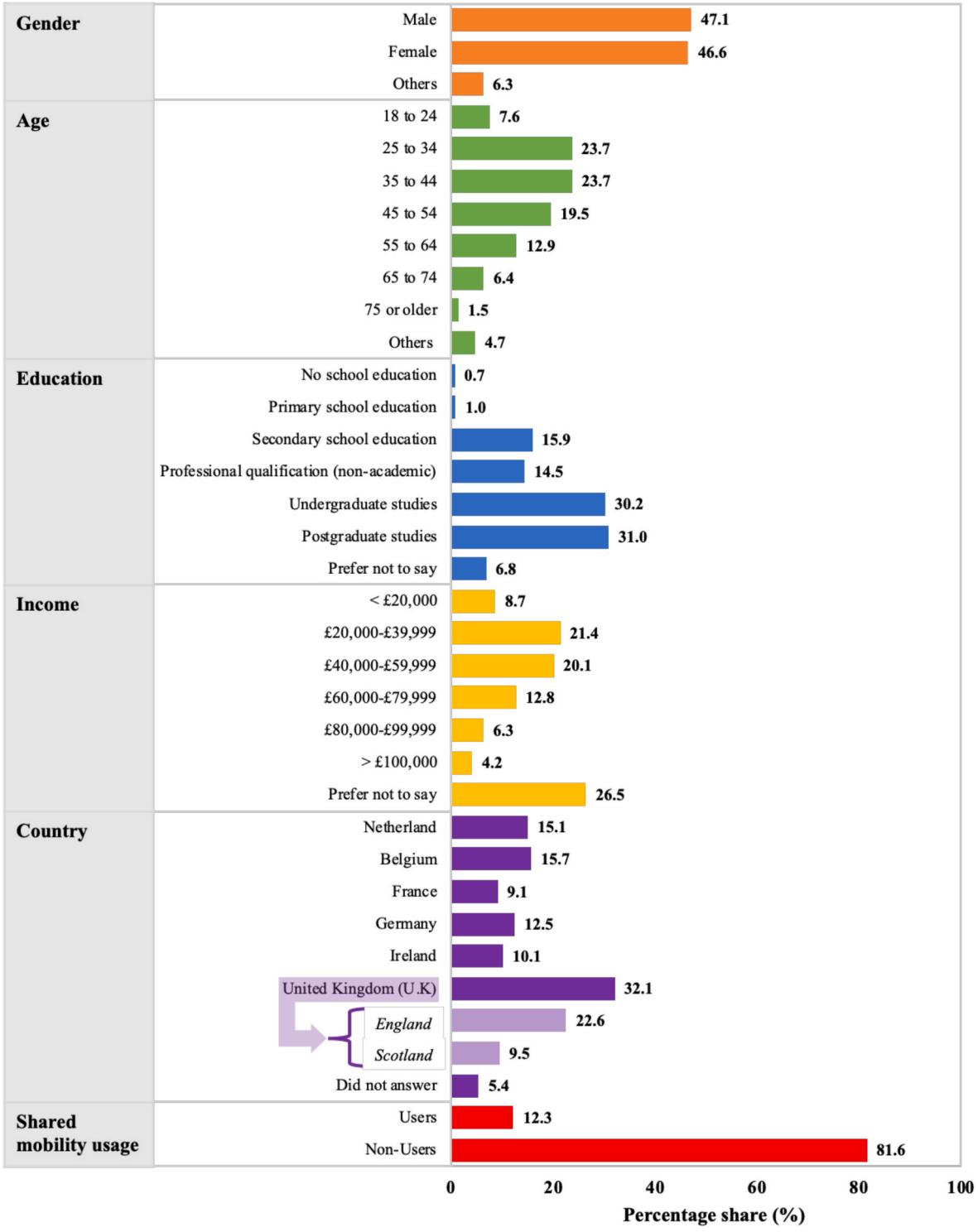


Fig. 1. Survey respondents characteristics.

SEM (PLS-SEM) does not require the data to obey the assumption of multivariate normality. In the present study, one-sample Kolmogorov-Smirnov (K-S) test performed on the collected data (using SPSS software) revealed that the K-S statistic values were in the range of 0.12–0.24 with a p-value <0.05 [Figure A1], indicating that the data did not conform to the assumption of multivariate normality. Considering the non-normal distribution of the data in our study, PLS-SEM was used in further analysis. PLS-SEM comprises two key components: the measurement model (or outer model) and the structural model (or inner model). The measurement model examines the relationship between the latent variable and its observed indicators (manifested variables), while the structural model examines the direct and indirect relationship between the latent variables. The measurement model is expressed by Equation (1):

$$x_k = \pi_{k0} + \pi_k \xi + \varepsilon_k \quad (1)$$

Where, x_k is the observed indicator; π_k indicates factor loading related to x_k; and is the construct to which x_k is associated, ε_k is the error term.

The structural model is expressed by Equation (2):

$$\xi_j = \beta_{j0} + \sum_i \beta_{ji}\xi_i + \nu_j \quad (2)$$

Where, ξ_j is the jth construct with total i construct, β is the regression coefficient of construct and ν_j is the error term.

In PLS-SEM, the path model estimation occurs in four stages (Henseler et al., 2016). In the first stage, composite scores for the latent variable or construct are determined by an iterative PLS regression algorithm. The second stage deals with the correction of attenuation for constructs modelled as factors. In the third stage, the parameters estimation is performed, which provides an estimate for loading, effects (indirect, total), and model assessment criteria. Finally, in the fourth stage, bootstrapping is performed to examine the significance of path relationship.

In the present study, the measurement model was evaluated first, followed by the evaluation of the structural model as suggested by Hair et al. (2017). Initially, reliability and validity tests were performed on the measurement model to check if the data was suitable for further analysis. Once the acceptable validity and reliability was achieved, the structural model was tested for model fit. In PLS-SEM, the model fitness can be evaluated based on standardized root mean square residual (SRMR), Normalized fit index (NFI), coefficient of determination (R^2), and predictive relevance (Q^2). In PLS-SEM, the R^2 value only indicates the explanatory power of the model, whereas the predictive power of the model is estimated by the Q^2 value, which is estimated using the blindfolding technique, where single data points are omitted, and replaced by the mean value and then PLS estimation is made (Galit et al., 2019). Finally, for comparing the participants belonging to different socio-demographic groups based on their intentions and attitudinal factors, non-parametric tests such as Mann Whitney U test (for comparing two independent groups) and the Kruskal-Wallis test (for comparing independent groups more than two) was performed (Simon et al., 2020). Fig. 2 illustrates the analytical framework used in the present study.

4. Results

In this section the results are presented systematically according to each step in the analysis described in the previous section and shown schematically in Fig. 2.

4.1. Principal Component Analysis (PCA)

As an initial step, PCA with oblique rotation (Promax) was performed. In line with past studies, only items with component loadings greater than 0.4 were considered (Hair, 2017). PCA revealed five components having eigenvalues greater than 1. However, the fifth component was not fulfilling the minimum criteria of internal consistency, therefore the PCA was again performed with four components. A Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy = 0.911 and statistically significant Bartlett's Test of Sphericity (BTS) value of (χ^2 (325) = 25962.8, $p < 0.0001$), indicated that the data was suitable for the analysis. Overall, the four components explained 50% of the total variance. Table 2 shows the results of the PCA performed on the questionnaire survey data.

The components were named according to the category of items they encompassed. Component 1 consisted of attitudinal items related to environmental belief of individuals towards car use, subjective and personal norms. Component 2 consisted of most of the items related to negative attitudes of individuals towards using shared mobility in terms of their current habits, car use, perceived compatibility and relative

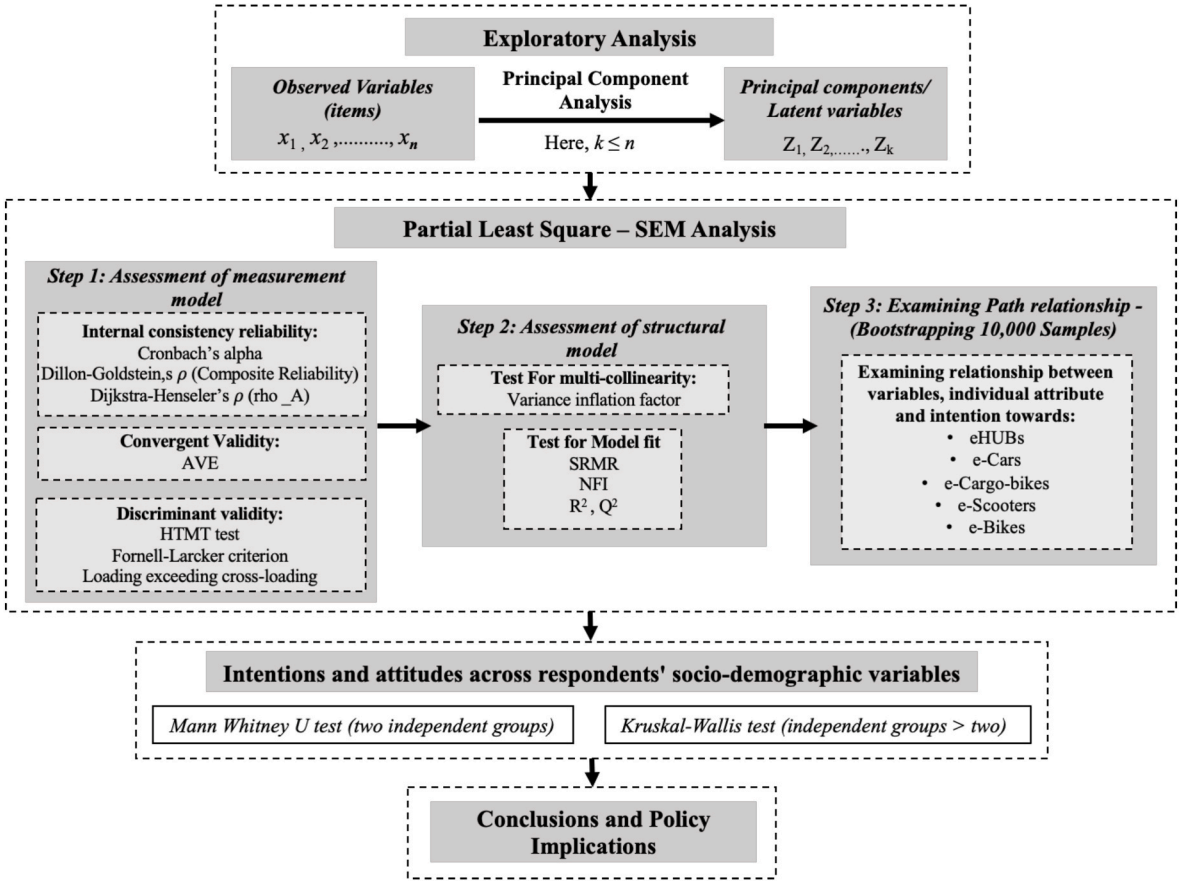


Fig. 2. Analytical framework used in the study.

Table 2
Results of PCA performed on questionnaire survey.

Items	Measured Construct	Component			
		1	2	3	4
Att18: I feel a moral obligation to reduce my emissions of greenhouse gases.	*Personal norms (TPB)*	0.823			
Att19: For the sake of the environment, everyone should reduce how much they use cars.	*Car use attitude (TPB)*	0.802			
CarAtt6: Next time I buy a car, I would consider buying a car with lower CO_2 emissions.	*Environmental attitude (TPB)*	0.704			
Att17: Congestion, air pollution and noise from road traffic is a real problem in my city.	*Environmental attitude (TPB)*	0.558			
Att20: People who drive cars that are better for the environment should pay less to use the roads.	*Car use attitude (TPB)*	0.570			
CarAtt1: I am willing to reduce the amount I travel by car.	*Car use attitude (TPB)*	0.448			
Att13: People around me find it important to reduce emissions of greenhouse gases.	*Perceived subjective norms (TPB)*	0.505			
Att3: I prefer travelling the way I'm used to rather than using eHUBs.	*Habit*		0.738		
Att9: There is no point in using shared mobility options if you already own a car.	*Relative advantage (DOI)*		0.735		
Att6: I'd rather wait for other people to try eHUBs before I use them.	*Trialability (DOI)*		0.753		
Att4: Shared mobility solutions like eHUBs are too complicated for me to use.	*Complexity (DOI)*		0.685		
Att10: Shared mobility options can't fulfil my mobility needs.	*Perceived compatibility (DOI)*		0.716		
Att16: People should be allowed to use their cars as much as they like, even if it causes damage to the environment.	*Car use attitude (TPB)*	−0.507	0.504		
CarAtt8: There is no point in reducing my car use to help the environment unless others do the same.	*Personal norms (TPB)*		0.461		
Att11: I do not feel confident to use an electric car.	*PBC E-cars (TPB)*		0.437		
Att7: I would enjoy trying out and using different electric vehicles from eHUBs.	*Trialability (DOI)*			0.668	
Att2: I'd be interested in using eHUBs for non-work trips when they've become available in my city.	*Adoption Intentions for leisure (TPB)*			0.633	
Att5: I am confident that, if I wanted to, I could use eHUBs without problems.	*PBC eHUBs (TPB)*			0.686	
Att15: I'm often among the first people to experiment with new technologies.	*Affinity to technology*			0.725	
Att1: I'd be interested in using eHUBs for commuting trips when they've become available in my city.	*Adoption intentions for commute (TPB)*			0.624	
Att8: Shared mobility options provide me with more flexibility in the way I travel.	*Relative advantage (DOI)*			0.580	
Att12: I feel confident to ride an electric bicycle.	*PBC e-bikes (TPB)*			0.544	
CarAtt3: Many of the journeys that I now make by car I could just as easily walk.	*Attitude toward using alternate mode (TPB)*				0.820
CarAtt5: Many of the journeys that I now make by car I could just as easily do by using public transport.	*Attitude toward using alternate mode (TPB)*				0.787
CarAtt4: Many of the journeys that I now make by car I could just as easily cycle.	*Attitude toward using alternate mode (TPB)*				0.770
CarAtt7: People around me (my family, friends and/or colleagues) encourage me to reduce my car use.	*Perceived subjective norms (TPB)*				0.525
No. of Items		**7**	**8**	**7**	**4**
Cronbach's alpha		**0.78**	**0.81**	**0.83**	**0.75**

advantage. Component 3 consisted of items which were closely related to individual positive attitude towards using shared mobility and finally Component 4 compromised most of the items which represented individual willingness to use alternate mode to replace their current car usage. Considering this scrutiny of the component characteristics, they were named as *Environmental belief, Positive attitude towards shared mobility, Negative attitude towards shared mobility, and Alternate mode acceptance.*

4.2. Partial least square SEM

Once latent variables were derived from the PCA, a conceptual model for car users' intention towards eHUBs was created which also included their socio-demographic variables [Fig. 3]. As suggested by Hair et al. (2017), the measurement model was assessed to examine the relationship between manifest and latent variables and validity and reliability tests were carried out on the measurement model. After establishing the

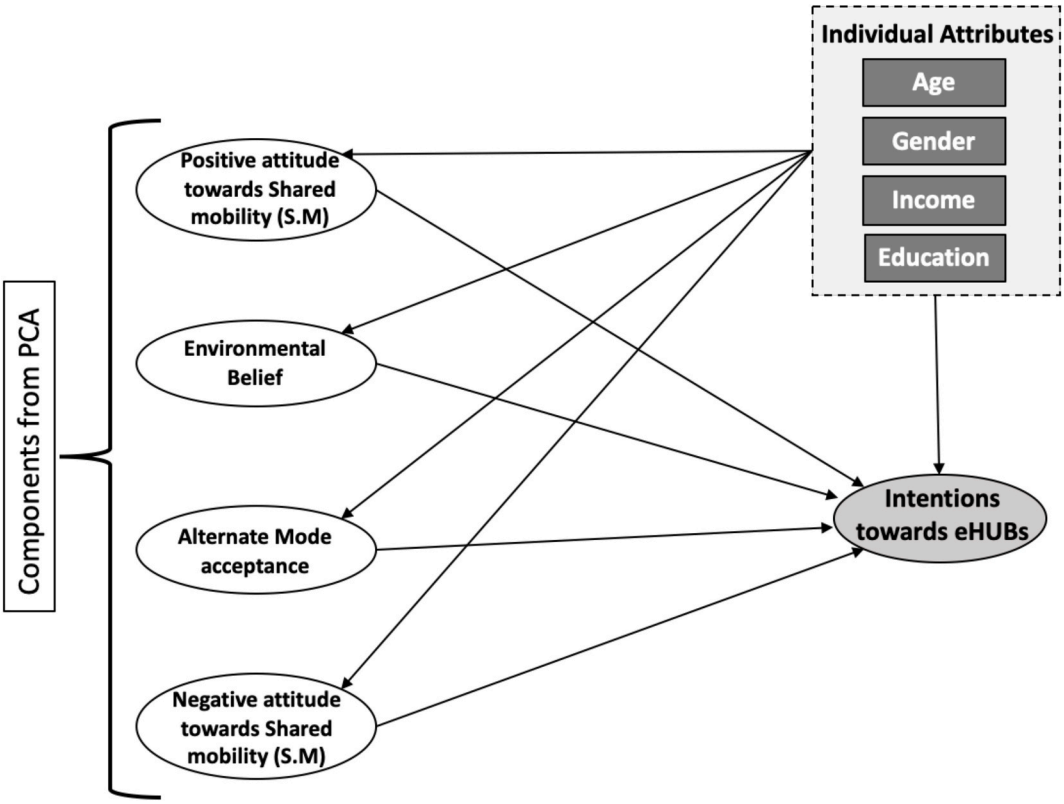


Fig. 3. Conceptual model for car user's intentions towards eHUBs.

validity and reliability of the measurement model, an assessment was conducted on the structural model to explore the relationship between the latent variables and car users' intention towards using eHUBs.

4.2.1. Assessment of measurement model

Initially, confirmatory factor analysis was performed to confirm and refine the items included in the measurement model. Henseler et al. (2016) suggested three criteria for evaluating measurement model: convergent validity, internal consistency reliability, and discriminant validity. To assess the convergent validity, the average variance extracted (AVE) and the factor loading of each observed variable with the latent variable were examined. A component loading $\geq$0.60 was considered to be significant (Hair et al., 2017; Nguyen-Phuoc et al., 2021). For all the latent variables the AVE was greater than 0.50, hence the convergent validity was adequate [Table 3].

The Cronbach's alpha value for each construct was evaluated to test the internal consistency among the observed variables within each construct. Table 3 indicates that Cronbach's alpha values for the latent variables exceeded 0.70. Furthermore, the composite reliability [Dillon-Goldstein's ρ and rho_A (Dijkstra Henseler's ρ)] for the latent variables also were greater than 0.70, which signifies that there exists good internal consistency among the constructs (Henseler et al., 2016).

For the assessment of the measurement model for discriminant validity, the Fornell-Larcker criterion was used. Table B1 illustrates that the

Table 3
Measurement model evaluation.

Latent Variable	Observed Variable	Component loading	Cronbach's alpha	Dillon-Goldstein's ρ	Dijkstra Henseler's ρ	AVE
Environmental Belief	Att19	0.803	0.78	0.79	0.85	0.53
	CarAtt1	0.763				
	Att18	0.803				
	Att17	0.634				
	Att13	0.623				
Negative Attitude (S.M)	Att3	0.854	0.78	0.85	0.84	0.53
	Att9	0.798				
	Att10	0.708				
	Att4	0.652				
	Att6	0.595				
Positive Attitude (S.M)	Att2	0.851	0.85	0.86	0.89	0.57
	Att7	0.832				
	Att8	0.762				
	Att1	0.801				
	Att5	0.658				
	Att12	0.611				
Alt. mode acceptance	CarAtt4	0.833	0.76	0.77	0.85	0.58
	CarAtt5	0.758				
	CarAtt3	0.739				
	CarAtt7	0.710				

square root of the Average Variance Extracted (AVE) for all constructs was greater than the square correlation with other constructs, demonstrating satisfactory discriminant validity of the constructs. Further, the cross-loading table reveals that all the observed variables had the highest loading with their respective construct only [Table B2] and the HeteroTrait-MonoTrait ratio of correlation (HTMT) values were found well below 0.77, which is less than the threshold value of 1 [Table B3], thus indicating acceptable discriminant validity (Henseler et al., 2016).

4.2.2. Assessment of structural model

Before assessing the structural model, the variance inflation factor (VIF) values for all the indicators and latent variables were examined to ensure that there is no issue of multicollinearity. The VIF values for all the indicators/manifested variables and latent variables were less than 2.53 and 1.83, respectively (Table C1). Thus, for both latent variables and indicators, the VIF value did not exceed the threshold value of 5, indicating no issue of multicollinearity (Henseler et al., 2016). The standardized root means square residual (SRMR) value for the structural models were found to be less than 0.08, and NFI values were greater than 0.80, indicating a good fit (Henseler et al., 2016) [Table 4]. Further, R^2 and Q^2 values were within the acceptable ranges (Falk and Miller, 1992; Henseler et al., 2016), signifying that the models have acceptable exploratory power and predictive relevance, respectively [Table 4]. As a final step of structural model evaluation, bootstrapping (10,000 sub-samples) was performed to check the significance and relevance of the path coefficients for different models.

4.2.3. Path relationship evaluation

The relationships between variables (direct and indirect) were examined using regression coefficients (β). The path relationship was deemed to be statistically significant at 5% level (i.e., t-value >1.96). Table 4 presents the direct path relationship between the latent variables; Table 5 presents the indirect path relationship between the latent variables and Table 6 presents the total effect of indirect variables on intentions.

General findings: The results revealed that individuals' portraying a positive attitude towards shared mobility and open to accepting alternative modes of their travel showed higher intention towards using shared vehicles from eHUBs. On the other hand, individuals who held a negative attitude towards shared mobility were found to be less likely to have the intention to use shared vehicles from eHUBs. In terms of socio-demographic variables, age was found to have a direct negative relationship with positive attitude towards shared mobility and acceptance to use alternate mode. In terms of indirect effect, it was observed that positive attitude towards shared mobility and acceptance to use alternate mode moderated the existing relationship between age and intentions, where age was found to be negatively associated with intentions towards using vehicles from eHUBs. Interestingly, education had no direct relationship with intentions for different shared e-vehicles offered by eHUBs. Finally, the results of indirect effect revealed that males showed higher intentions towards using shared e-vehicles from eHUBs compared to females.

eHUBs (all modes): Regarding car user attitude towards eHUBs (all vehicles), it was found that positive attitude towards shared mobility ($\beta = 0.49$), negative attitude towards shared mobility ($\beta = -0.15$) and alternate mode acceptance ($\beta = 0.19$) was significantly associated with intentions, whereas environmental belief was not significantly associated with intentions to use shared mobility alternatives from eHUBs. In terms of socio-demographic variables, age ($\beta = -0.15$) and income ($\beta = -0.05$) was found to be negatively associated with intentions, whereas education has no significant effect on intention *(direct effect)*. Whereas indirect effect reveals that education was positively associated with intentions ($\beta = 0.79$).

Shared e-cars: For shared e-cars, socio-demographic variables such as age, gender, income, and education were not statistically significant predictors of intentions (*direct effect*). However, age was found to be negatively associated with alternate mode acceptance behaviour ($\beta = -0.17$) and positive attitude towards shared mobility ($\beta = -0.16$), this in turn has an indirect effect on intentions. Age indirectly affected intentions negatively ($\beta = -0.12$). Similarly, in terms of gender, males have higher intentions towards using shared e-cars ($\beta = 0.12$) (*indirect effect*) as they hold higher positive attitudes towards shared mobility ($\beta = 0.15$) and alternative mode acceptance behaviour ($\beta = 0.13$). Furthermore, the intentions to use shared e-cars was higher among respondents with higher educational qualification ($\beta = 0.08$) (*indirect effect*) because higher educational qualifications were positively associated with positive attitude towards shared mobility ($\beta = 0.15$) and

Table 4
Results of path relationship in structural model (*Direct effect*).

Path Relationship (Direct Effect)	All modes	e-Cars	e-Cargo-bike	e-Scooters	e-Bike
Negative Attitude (S.M) → Intention	**−0.149**	**−0.128**	**−0.127**	**−0.092**	**−0.146**
Positive Attitude (S.M) → Intention	**0.489**	**0.477**	**0.364**	**0.336**	**0.431**
Alt. mode acceptance → Intention	**0.192**	**0.103**	**0.176**	**0.172**	**0.202**
Environmental Belief → Intention	−0.027	−0.043	−0.007	**−0.054**	−0.030
Age → ntention	**−0.149**	−0.028	**−0.114**	**−0.248**	**−0.103**
Age → Alt. mode acceptance	**−0.171**	**−0.170**	**−0.168**	**−0.167**	**−0.168**
Age → Environmental Belief	**−0.083**	−0.032	**−0.076**	**−0.079**	−0.035
Age → Negative Attitude (S.M)	**0.083**	0.028	**0.078**	**0.079**	0.029
Age → Positive Attitude (S.M)	**−0.199**	**−0.161**	**−0.196**	**−0.197**	**−0.153**
Education → Intention	0.006	0.005	−0.016	−0.022	−0.023
Education → Alt. mode acceptance	−0.024	**−0.058**	−0.025	−0.031	**−0.056**
Education → Environmental Belief	**0.166**	**0.194**	**0.167**	**0.169**	**0.192**
Education → Negative Attitude (S.M)	**−0.135**	**−0.141**	**−0.134**	**−0.135**	**−0.144**
Education → Positive Attitude (S.M)	**0.128**	**0.146**	**0.126**	**0.130**	**0.152**
Gender → Intention	**0.083**	0.008	0.010	**0.130**	−0.003
Gender → Alt. mode acceptance	**0.142**	**0.132**	**0.138**	**0.125**	**0.137**
Gender → Environmental Belief	**−0.107**	**−0.142**	−0.102	**−0.108**	**−0.137**
Gender → Negative Attitude (S.M)	**−0.267**	**−0.165**	**−0.263**	**−0.264**	**−0.161**
Gender → Positive Attitude (S.M)	**0.269**	**0.155**	**0.274**	**0.267**	**0.186**
Income → Intention	**−0.052**	−0.030	−0.043	**−0.045**	−0.026
Income → Alt. mode acceptance	−0.031	−0.023	−0.028	−0.033	−0.021
Income → Environmental Belief	0.047	0.040	0.050	0.046	0.041
Income → Negative Attitude (S.M)	−0.029	−0.017	−0.027	−0.032	−0.020
Income → Positive Attitude (S.M)	0.023	0.000	0.027	0.024	0.009
Model Fit:					
SRMR	0.066	0.067	0.066	0.065	0.067
NFI	0.808	0.800	0.807	0.814	0.802
R^2 (Intention)	0.502	0.325	0.312	0.321	0.376
Q^2 Predict (Intention)	0.083	0.016	0.045	0.111	0.038

Significant values are emphasized in bold.

Table 5
Results of path relationship in structural model (*specific indirect effect*).

Path Relationship (Specific indirect effect)		All modes	e-Cars	e-Cargo-bike	e- Scooters	e-Bike
Age	Negative Attitude (S.M) → Intention	**−0.012**	−0.004	**−0.010**	**−0.007**	−0.004
Age	Environmental Belief → Intention	0.002	0.001	0.001	0.004	0.001
Age	Positive Attitude (S.M) → Intention	**−0.097**	**−0.077**	**−0.071**	**−0.066**	**−0.066**
Age	Alt. mode acceptance → Intention	**−0.033**	**−0.017**	**−0.030**	**−0.029**	**−0.034**
Education	Negative Attitude (S.M) → Intention	**0.020**	**0.018**	**0.017**	**0.012**	**0.021**
Education	Environmental Belief → Intention	−0.005	−0.008	−0.001	−0.009	−0.006
Education	Positive Attitude (S.M) → Intention	**0.063**	**0.069**	**0.046**	**0.044**	**0.066**
Education	Alt. mode acceptance → Intention	−0.005	**−0.006**	−0.004	−0.005	**−0.011**
Gender	Negative Attitude (S.M) → Intention	**0.040**	**0.021**	**0.033**	**0.024**	**0.024**
Gender	Environmental Belief → Intention	0.003	0.006	0.001	0.006	0.004
Gender	Positive Attitude (S.M) → Intention	**0.131**	**0.074**	**0.099**	**0.090**	**0.080**
Gender	Alt. mode acceptance → Intention	**0.027**	**0.014**	**0.024**	**0.022**	**0.028**
Income	Negative Attitude (S.M) → Intention	0.004	0.002	0.003	0.003	0.003
Income	Environmental Belief → Intention	−0.001	−0.002	0.000	−0.002	−0.001
Income	Positive Attitude (S.M) → Intention	0.011	0.000	0.010	0.008	0.004
Income	Alt. mode acceptance → Intention	−0.006	−0.002	−0.005	−0.006	−0.004

Significant values are emphasized in bold.

Table 6
Results of path relationship in structural model (*total indirect effect*).

Path Relationship (Total effect)	All modes	e-Cars	e-Cargo-bike	e-Scooters
Age → Intention	**−0.289**	**−0.125**	**−0.225**	**−0.346**
Education → Intention	**0.079**	**0.079**	0.041	0.019
Gender → Intention	**0.284**	**0.123**	**0.168**	**0.271**
Income → Intention	−0.044	−0.032	−0.035	−0.042

Significant values are emphasized in bold.

negatively associated with negative attitude towards shared mobility ($\beta = -0.14$). However, income had no direct or indirect influence on intentions towards using shared e-cars.

Shared e-cargobikes: For shared e-cargobike, results indicated that age was negatively associated with intentions towards using shared e-cargobikes ($\beta = -0.11$) (*direct effect*), whereas education, gender and income have no statistically significant direct effect on intentions. However, in terms of indirect effect, compared to females, males showed higher intentions towards using shared e-cargobike ($\beta = 0.17$) as they had higher positive attitude ($\beta = 0.27$), higher alternate mode acceptance behaviour ($\beta = 0.14$) and lower negative attitude toward shared mobility ($\beta = -0.26$). In addition, income and education showed no direct or indirect effect on the intentions to use shared e-cargobike.

Shared e-scooters: Similar to shared e-cargobikes, age was negatively associated ($\beta = -0.25$) with intentions towards using shared e-scooters. The intention towards using shared e-scooters was also higher among male than females ($\beta = 0.13$). Furthermore, higher income was associated with lower intentions towards using shared e-scooters ($\beta = -0.04$) and education had no statistical significance either direct or indirect effect on intentions towards shared e-scooters.

Shared e-bike: Intentions to use shared e-bike was found to be negatively associated with age ($\beta = -0.10$) whilst other socio-demographic variables such as gender, income and education had no direct influence on shared e-bike intentions. In terms of indirect effects, compared to females, males showed higher intentions ($\beta = 0.13$) in

Table 7
Mean scores of intentions across socio-demographic variables.

Socio-demographic variable	Categories	Intentions [Mean (St. Dev.)]			
		e-Cars	e-Cargo-bike	e- Scooters	e-Bike
Age	**18 to 24**	58.9 (31.8)	42.2 (33.4)	53.1 (33.3)	56.7 (30.3)
	25 to 34	54.8 (33.4)	38.4 (34.3)	42.1 (34.2)	50.7 (33.4)
	35 to 44	49.0 (34.8)	33 (33.8)	32.8 (34.6)	43.2 (35.1)
	45 to 54	45.3 (35.8)	24.7 (31.5)	22.6 (30.2)	37.6 (35.1)
	55 to 64	43.3 (36.1)	20.1 (29.5)	15.7 (26.5)	35.8 (35.6)
	65 to 74	37.4 (35.1)	12.8 (23.1)	11.2 (22.3)	28.6 (32.9)
	75 or older	38.9 (33.2)	19.1 (25.3)	13 (22.2)	25.6 (32.4)
	H(6)	**86.15**[a]	**148.79**[a]	**278.34**[a]	**155.12**[a]
Gender	**Female**	46.6 (35.1)	28.3 (32.2)	27.8 (32.4)	40.7 (35)
	Male	50.7 (34.9)	33 (34.2)	35.2 (35.4)	44.4 (35.1)
	Z	**−3.38**[a]	**−3.14****	**−4.76**[a]	**−2.97**[a]
Education	**No school education**	50.1 (34.9)	38.6 (29.3)	36.4 (28.5)	61.3 (31.2)
	Primary school education	47.4 (37.9)	40.5 (36.5)	32.5 (34.6)	41.9 (33.8)
	Secondary school education	42.4 (35.4)	26 (32.2)	27.1 (33.0)	36.4 (35)
	Professional qualification (non-academic)	48.5 (34.7)	28.9 (33.1)	30.4 (34.3)	44.1 (35.9)
	Undergraduate studies	49.8 (34.7)	32.6 (33.2)	32 (32.8)	43.3 (34.3)
	Postgraduate studies	51.1 (34.9)	31.9 (34.2)	33.6 (35.9)	44.1 (35.2)
	H(5)	**24.99**[a]	**20.96**[a]	**12.07**[a]	**30.61**[a]
Income	**< £20,000**	52 (34.2)	37.5 (34.3)	38.4 (36.9)	47.7 (34.9)
	£20,000-£39,999	57.5 (32.3)	36.1 (33.9)	37.8 (35.2)	49.4 (33.8)
	£40,000-£59,999	53.7 (32.7)	36.3 (34)	36.4 (33.3)	47.1 (34.4)
	£60,000-£79,999	54.0 (32.4)	35 (32.5)	35.4 (31.8)	46.4 (32.8)
	£80,000-£99,999	58.8 (31.7)	31.6 (31.8)	32.9 (32.3)	49.4 (34.9)
	> £100,000	52.9 (33.7)	38.4 (35.9)	41.1 (35.2)	49.2 (34.1)
	H(5)	10.03	2.40	3.40	3.19
	Overall Sample	47.3 (35.4)	29.3 (33)	30 (33.7)	41.5 (35.2)

[a] $p < 0.05$, H = Kruskal-Wallis H-value; Z = Mann-Whitney *U* test Z-Score.

terms of using shared e-bikes as they had a higher positive attitude ($\beta = 0.19$), higher alternate mode acceptance behaviour ($\beta = 0.14$), and a lower negative attitude toward shared mobility ($\beta = -0.16$). Also, educational background had a statistically significant positive influence ($\beta = 0.05$) (*indirect effect*) on the intention to use shared e-bikes.

4.3. Intentions and attitudes across respondents' socio-demographic variables

For participants belonging to different socio-demographic categories, Table 7 presents the mean scores of intentions for different vehicles offered by eHUBs and Table 8 presents the mean scores across attitudinal factors. In general, the results indicate that people showed higher intentions towards using shared e-cars and e-bike whereas intentions towards using e-cargobike and e-scooters remained lower [Table 7]. For all the shared vehicles offered by eHUBs, younger age participants showed higher intentions towards using different vehicles than older age participants. For participants younger than 24 years, e-cars, e-bikes, and e-scooters were the most favorable mode, whereas the intentions for e-cargobike remained comparatively low. In terms of gender, males showed higher intentions towards using different vehicles offered by eHUBs compared to females and there were no statistically significant differences between participants belonging to different income groups in terms of their intentions.

With respect to attitude, younger participants showed higher positive attitude towards shared mobility, environmental belief and showed greater interest to use alternate modes for their travel [Table 8]. In contrast, older participants showed a higher negative attitude towards shared mobility and showed less interest in using alternative modes for their travel. In terms of gender, female participants had higher environmental belief and showed a higher negative attitude towards shared mobility than males. Male participants, on the other hand, showed a higher positive attitude towards shared mobility and showed greater interest towards using alternative modes for their travel. In addition, environmental belief and positive attitudes were higher among those with postgraduate educational qualification and those receiving higher incomes.

5. Discussion

The present study analyzed car users' intentions towards using different vehicles offered by eHUBs. Firstly, the study examined how participants' attitudes towards shared mobility and their car use attitudes are associated with their intentions towards using different vehicles offered by eHUBs. Secondly, the study examined the effect of individual socio-demographic variables on their attitudes and intentions towards using shared e-vehicles. Overall, the findings of the present study advance research in the area of shared mobility where car users' intentions towards multi-modal shared mobility schemes such as eHUBs is examined, which is in contrast to studies available in the literature (Haustein and Jensen, 2018; Kim et al., 2015; Mitra and Hess, 2021; Mouratidis, 2022) where users' intentions are only examined for unimodal vehicle sharing schemes offering only one type of vehicle for their travel.

The findings indicate that attitude towards shared mobility emerged as a significant predictor of car users' intentions towards using shared vehicles from multi-modal vehicle sharing schemes. Those having a higher positive attitude towards shared mobility showed higher intentions to use shared e-cars, e-cargobikes, e-scooters and e-bikes. These were the individuals who would enjoy and feel confident in using shared e-vehicles from eHUBs and believed that shared mobility has potential benefits in terms of providing them with more flexibility for their travel requirements. Similar findings have been reported by Jie et al. (2021) for the adoption of shared mobility in Australia, Li et al. (2021) for shared e-bikes in China, and Oliveira et al., 2022 for shared electric cars in Brazil. In contrast, individuals with higher negative attitudes towards shared mobility evidenced lower intentions to use different shared vehicles offered by eHUBs. Furthermore, individuals with higher acceptance towards using alternative modes for their travel needs showed higher intentions towards using shared vehicles from eHUBs. This suggests that individuals who believed that other sustainable travel modes can substitute their current car use also showed higher intentions to use

Table 8
Differences in attitudes across socio-demographic variables.

Socio-demographic variable	Categories	Mean score [Mean (St. Dev)]			
		Environmental Belief	Negative Attitude (S. M)	Positive Attitude (S. M)	Alt. mode acceptance
Age	**18 to 24**	4.73 (1.06)	3.89 (1.21)	4.57 (1.13)	3.24 (1.26)
	25 to 34	4.91 (1.11)	3.81 (1.31)	4.62 (1.27)	3.07 (1.37)
	35 to 44	4.86 (1.17)	3.76 (1.30)	4.50 (1.33)	2.78 (1.30)
	45 to 54	4.86 (1.19)	3.73 (1.33)	4.36 (1.40)	2.64 (1.29)
	55 to 64	4.78 (1.26)	3.87 (1.41)	4.28 (1.47)	2.54 (1.21)
	65 to 74	4.57 (1.22)	4.07 (1.41)	3.81 (1.53)	2.62 (1.14)
	75 or older	4.09 (1.08)	4.47 (1.38)	3.17 (1.35)	2.23 (0.96)
	H(6)	**41.33**[a]	**25.56**[a]	**87.40**[a]	**101.44**[a]
Gender	**Female**	4.88 (1.11)	3.94 (1.33)	4.29 (1.37)	2.75 (1.29)
	Male	4.76 (1.23)	3.70 (1.32)	4.54 (1.36)	2.86 (1.31)
	Z	**−2.40**[a]	**−5.33**[a]	**−5.21**[a]	**−2.23**[a]
Education	**No school education**	4.26 (0.94)	3.74 (1.20)	3.61 (1.25)	3.04 (1.42)
	Primary school education	4.03 (1.02)	3.52 (1.28)	3.71 (1.47)	3.23 (1.09)
	Secondary school education	4.45 (1.19)	4.25 (1.27)	3.97 (1.48)	2.86 (1.40)
	Professional qualification (non-academic)	4.68 (1.21)	3.95 (1.27)	4.38 (1.37)	2.92 (1.40)
	Undergraduate studies	4.86 (1.13)	3.76 (1.37)	4.48 (1.36)	2.76 (1.23)
	Postgraduate studies	5.10 (1.11)	3.59 (1.31)	4.65 (1.25)	2.75 (1.28)
	H(5)	**143.62**[a]	**96.89**[a]	**88.90**[a]	10.37
Income	**< £20,000**	4.66 (1.19)	4.00 (1.33)	4.19 (1.43)	2.89 (1.28)
	£20,000-£39,999	4.79 (1.12)	3.83 (1.30)	4.39 (1.35)	2.91 (1.35)
	£40,000-£59,999	4.87 (1.14)	3.83 (1.33)	4.46 (1.33)	2.84 (1.28)
	£60,000-£79,999	4.96 (1.10)	3.71 (1.28)	4.48 (1.21)	2.84 (1.19)
	£80,000-£99,999	4.93 (1.26)	3.84 (1.38)	4.60 (1.41)	3.04 (1.55)
	> £100,000	4.95 (1.23)	3.70 (1.28)	4.49 (1.33)	2.72 (1.33)
	H(5)	**18.56**[a]	10.15	**13.17**[a]	3.87

[a] $p < 0.05$, H = Kruskal-Wallis H-value; Z = Mann-Whitney *U* test Z-Score.

shared vehicles from eHUBs. However, in contrast to the findings of Kopplin et al. (2021) and Li et al. (2021), the findings of the present study reveal that environmental belief does not have any statistically significant influence on car user's intentions towards using shared vehicles from multi-modal vehicle sharing schemes. In sum, the findings suggest that car users' intentions toward using shared mobility are significantly affected by their attitude towards shared mobility (positive or negative) and their acceptance towards using alternate modes for their travel requirements rather than their environmental beliefs.

In terms of demographic variables, age emerged as a statistically significant predictor of individual attitudes and intentions towards using shared e-vehicles. Supporting previous findings of Bösehans et al. (2023a), Curtale et al. (2021), and Mitra and Hess (2021), the findings of the present study reveal that intentions towards using shared e-vehicles were higher among younger participants, whilst older age participants showed lower intentions towards using shared e-vehicles. This is supported by the finding that younger age participants had higher environmental belief, a higher positive attitude towards shared mobility, and higher acceptance towards substituting their current car use behaviour by other more sustainable modes. Older participants, on the other hand, held higher negative attitudes towards shared mobility and had less acceptance towards using any other mode for their travel requirement instead of their own private cars. Butler et al. (2021) also observed that older people are less inclined to use shared mobility services because they are less familiar with smartphones, online route planning, and ride-sharing applications. Furthermore, their habitual use of private vehicles is psychologically ingrained, making it difficult for them to adopt shared mobility. In respect to gender, consistent with the findings of Curtale et al. (2021), and Dorner and Berger (2020) it was found that males had higher intentions towards using shared e-vehicles than females. In terms of attitude, females had higher environmental belief, whereas males had a higher positive attitude towards shared mobility and showed higher acceptance towards using alternate modes for their travel. Furthermore, the effect of education was statistically significant only for shared e-cars and e-bikes, where individuals possessing higher educational qualifications had higher intentions towards using shared e-cars and e-bikes. In terms of attitudes, individuals with higher education qualifications evidenced higher environmental belief and a higher positive attitude towards shared mobility. However, in contrast to the findings of Jie et al. (2021), and Kim et al. (2015), in the present study income showed no significant influence on the intention to use shared e-vehicles. A possible explanation for the lack of an effect of income on intention could be that the sample consisted solely of car users. Car users might be less cost-sensitive overall, as owning a car already incurs high costs (i.e., fuel, insurance, and maintenance). As a consequence, the cost of shared mobility options may play a subordinate role for car users, who might prioritize other aspects such as convenience and comfort. Overall, the comparison of mean scores shows that people showed higher intentions towards using shared e-cars and e-bikes, whereas intentions towards e-cargobike remained the lowest among all the vehicles offered by eHUBs. Finally, the findings also reveal that older participants showed higher intentions towards using shared e-cars in preference to any other mode offered by eHUBs.

6. Policy implications

The findings of the present study reveal that different vehicles offered by eHUBs appeal to the different groups of car users. This in turn provides substantial evidence to support the benefits of multi-modal vehicle sharing scheme over unimodal vehicle sharing scheme. Fig. 4 shows the derived importance of each of the attitudinal variables and their relative performance. Here, the derived importance is based on the standardized path coefficient of each of the attitudinal factors with the intentions, whereas the performance of each of the attitudinal factors is calculated by the performance index given by Equation (3).

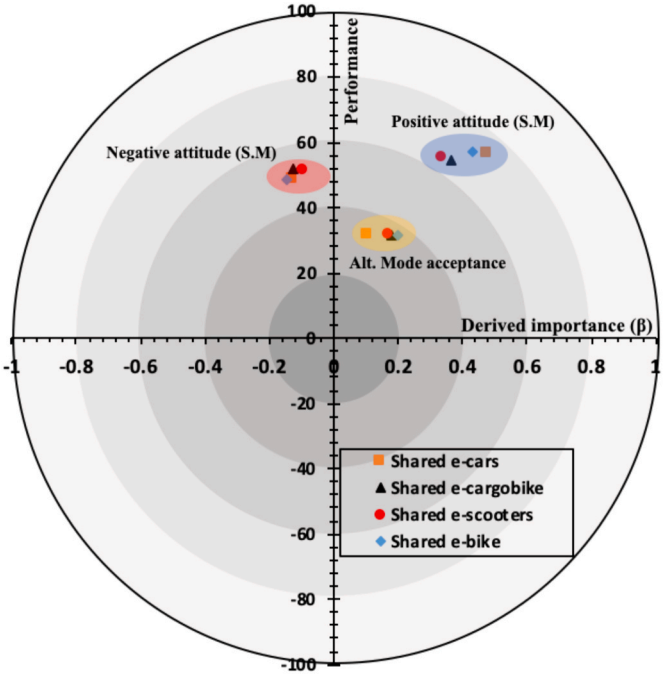


Fig. 4. Derived importance and performance of variables for different shared e-vehicles.

$$P.I = \frac{\sum_{i=1}^{k} \beta_i \overline{y_i} - \sum_{i=1}^{k} \beta_i}{(r-1)\sum_{i=1}^{k} \beta_i} \cdot 100 \quad (3)$$

Where, β_i is the weight of indicator i measured for a service dimension; $\overline{y_i}$ is the mean value of indicator i; k is the number of indicators in a service dimension, and r is the Likert scale value used in the study.

Fig. 3 shows that for car users, a positive attitude towards shared mobility and acceptance to use alternate modes for their travel exert a statistically significant positive influence on their intentions to use shared e-vehicles. In contrast, a negative attitude towards shared mobility acts as a barrier for car users, which prevents them from the uptake of shared mobility. For shared mobility providers, it is very important to understand that in order to motivate car users to avoid using their private cars and instead use shared mobility, they have to significantly improve the car users' positive attitude towards shared mobility and their acceptance towards using alternative sustainable transport modes for their travel. However, this is not an easy task, as car users tend to exhibit car stickiness and prefer the car even in situations where other modes are superior (Innocenti et al., 2013). Moreover, previous research has suggested that car users react negatively to coercive measures and are more easily persuaded by facilitative or pull measures (Stradling et al., 2000), which require voluntary changes in behaviour. Approaches, such as self-regulation theory, may help car users to reduce car use or even abandon their car altogether, but a prerequisite for such change to occur is an initial desire to reduce car use and to set corresponding goals (Bamberg et al., 2011). Hence, demonstrating how shared mobility options can supplement or substitute the car (e.g., through tailored applications) is crucial. For example, Zhang et al. (2022) emphasized that policymakers and platforms should highlight how private vehicle owners' behaviours can significantly impact traffic congestion and reduce environmental pollution. They recommended promoting the benefits and noticeable characteristics of ridesharing to foster a sense of responsibility among drivers and reshape their attitudes. Bamberg et al. (2011) highlighted several techniques to make social norms more salient and enhance the sense of responsibility,

including mass media role modelling, scenario-based risk information, and raising awareness through conscience-raising activities. Moreover, Bamberg et al. (2011) suggest that providing information about alternative travel options, along with evaluations of their benefits for example time saving will be most effective for car users who have already committed to reducing their car usage. However, Kreemers et al. (2021) pointed out that car users are often satisfied with their private vehicles when they meet mobility needs, and often ignore information about alternative modes due to attention bias. To overcome this, Kreemers et al. (2021) proposed targeting users based on their identity and engaging them when they are more open to alternatives, such as during car repairs or in relevant locations like parking spots.

In the present study, younger participants showed a higher positive attitude towards shared mobility and showed greater interest in replacing their current private car trips with other sustainable modes. However, older respondents showed higher negative attitudes towards shared mobility, where they highly believe that shared mobility is too complicated to use, and shared mobility cannot fulfil their mobility needs. It is important to understand that older people have considerably less experience with smartphones and the use of online applications to book a shared ride or vehicle (Butler et al., 2021). Furthermore, previous research has shown huge differences in older people's travel habits, with some evidencing car dependency, while others travel actively in a multi-modal fashion or are restricted in their mobility (Haustein and Siren, 2015). Importantly, a major deterrent to using shared mobility for older people and those with disabilities is the lack of suitable vehicle and service designs (Goralzik et al., 2022). However, as older people and those with disabilities are not primary target groups for shared mobility providers, therefore there is a lack of research interest in their specific shared mobility needs. Nevertheless, we provide tentative suggestions on how some of the shared mobility needs of the elderly could be addressed. For the older people who habitually use cars and are less familiar with technology, shared mobility providers should focus on creating simple, user-friendly smartphone applications rather than using complex applications for booking the vehicles. Harvey et al. (2019) also highlighted that smartphone application complexity and over-functionality often deters older people and suggested prioritizing basic functionality, hiding advanced features unless needed. Additionally, offering weekly or monthly subscriptions could simplify payments for older people, as these payment models are often found to enhance convenience and encourage greater adoption (Butler et al., 2021; Feneri et al., 2022). Also, studies (Hensher et al., 2022; Kreemers et al., 2021) have shown a higher affinity towards using a shared vehicle among users with prior experience. Thus, it would be beneficial to offer introductory discounts or free use to the older people for initial ride(s), so that they obtain experience with the use of shared e-vehicles. Given the higher intentions among older participants towards using shared e-cars than any other modes offered by eHUBs, it is crucial for shared mobility providers to offer facilities and vehicles adapted to meet their unique mobility needs.

The findings also indicate that female participants exhibited lower intentions to utilize e-shared mobility alternatives compared to male participants and demonstrated a lower positive attitude towards shared mobility and a lesser acceptance of alternate modes for their travel than males. A recent study by Catulli and Buckland-Stubbs (2024) suggests that "women have unequal access to mobility offerings and have differing transport needs and transport challenges to those of men, including factors related to safety concerns, convenience, cost and comfort". The authors suggest that, to enhance the uptake of MaaS alternatives among females, several strategies can be employed, ranging from improving infrastructure and app design to improving service delivery by taking a collaborative approach to how transport offerings are designed and implemented (Catulli and Buckland-Stubbs, 2024). Gorrini et al. (2021) emphasized addressing gender equity in shared mobility by proposing measures such as ensuring good visibility, adequate lighting, and CCTV monitoring at stations, alongside collaboration among stakeholders and law enforcement for effective monitoring and follow-up of issues to enhance the sense of security among female users. The authors also emphasized the importance of designing the shared mobility system that addresses diverse travel requirements of females, such as optimizing station locations (e.g. near schools, nurseries, shopping malls) and adapting vehicle design to better accommodate caregiving and domestic responsibilities (Gorrini et al., 2021).

In general, shared mobility providers and local authorities should work in close collaboration with each other to provide the facilities, vehicles and the incentives to motivate people to reduce their private car use and improve the uptake of shared mobility. In order to reduce negative attitudes and promote the positive attitude towards shared mobility, people should be made aware of the potential benefits to the environment and the cost of owning a vehicle and its maintenance. This can be achieved via the mobile phone application itself, by informing the user of their carbon emissions savings or the health benefit of being active after each of their rides. Furthermore, media campaigns and advertisements also play an essential role in raising the awareness among car users by highlighting the potential benefits of using shared vehicles, over the use of privately owned vehicles. Finally, local authorities should consider restricting private car use by reducing the number of private vehicle parking spaces simultaneously with prioritizing policies to provide and promote multi-modal shared vehicle hubs in cities (Bösehans et al., 2023a; Kim et al., 2017). Moreover, implementing congestion pricing for private vehicles also can promote car users' willingness to shift from using their private vehicles and instead use shared mobility alternatives (Burghard and Scherrer, 2022).

7. Conclusions and limitations

The findings of the present study indicate that there is much potential for multi-modal e-vehicle sharing schemes where car users have reported their intentions to use different vehicles for their travel. The wide range of vehicles offered by eHUBs appeal to different groups of car users to meet their travel needs. Specifically, car users showed the highest positive attitude towards the use of shared electric cars, thus indicating a tendency towards car-centric options. In terms of emission reduction goals, this tendency will only prove beneficial if shared e-mobility leads to reductions in car ownership, as also suggested by (Chapman et al., 2020). Hence, as car-centric options are generally preferred by current car drivers (see also (Bösehans et al., 2023a, 2023b), it is crucial to make shared e-car options easily accessible, widely available and as affordable as possible, so as to outweigh the benefits of private car ownership. Our findings also indicated that there exist certain barriers which restrict the adoption of shared mobility among car users, especially older respondents. Lack of awareness, negative attitudes, perceived difficulty and lack of confidence among car users are some of the barriers restricting their use of shared e-vehicles. Findings suggest that there is a significant need for shared mobility providers and local authorities to work in close collaboration with each other to motivate people to use shared mobility alternatives rather than owning private cars. This in turn has a positive impact on the environment and ameliorates the problem of traffic congestion and reduces the need for parking in the cities.

The present study has certain limitations, which could nevertheless guide future studies. As one of the common issues with online surveys, such as self-selection bias also apply to the current study. It is possible, for instance, that survey respondents with a greater interest in new technologies were more eager to participate in the survey. To some extent, through the use of online panels in Amsterdam and Manchester, issues of self-selection bias could be somewhat mitigated though, through survey quotas set by the authors. Also, it should be noted that respondents who are 65 years or older are underrepresented in the data, particularly in cities where data collection occurred via the cities' own distribution channels. Online surveys tend to exclude a sizable proportion of the elderly population who are more easily reached via

traditional pen and paper questionnaires (Kelfve et al., 2020). This is important because older people could be one of the major beneficiaries of shared mobility. In practice, however, older people are not only less likely to drive, but even less likely to use shared mobility (Mouratidis, 2022), which is why they are often completely neglected as a potential customer base by shared mobility providers. Adequately capturing this group in future shared mobility studies therefore is crucial.

Also worthy of note is that the survey did not include any specific items that relate to the relative cost of using shared electric vehicles versus the private car. Through the omission of such cost-specific items, it was not possible to determine with certainty whether the cost of shared mobility options is important to car users of different income groups. Furthermore, the present study does not consider the differences in car user's perception across various pilot regions of the eHUBs project. While cultural differences were not a primary focus of this research (see Kavta et al. (2024), for the spatial transferability of choice models to different geographic contexts), it can be expected that in car-centric cultures, such as Germany and the UK, shared e-car options will be preferred to non-car shared alternatives, whereas the latter will find more acceptance in less car-centric, multi-modal countries, such as Belgium and the Netherlands. In general, however, there is substantial overlap with regard to the early adopters of shared mobility options between different cultural contexts (see Bösehans et al. (2023b), for an example). Finally, the findings of present study are geographically limited to pilot regions in European countries, thus the findings of the present study cannot be generalized for other regions, however, the analytical approach is transferable.

CRediT authorship contribution statement

Harpreet Singh: Writing – original draft, Methodology, Formal analysis, Conceptualization. **Ankit Kathuria:** Writing – review & editing, Methodology. **Kuldeep Kavta:** Writing – review & editing, Methodology. **Gustav Bosehans:** Writing – review & editing, Methodology. **Margaret Carol Bell:** Writing – review & editing, Supervision, Methodology. **Dilum Dissanayake:** Writing – review & editing, Methodology, Funding acquisition.

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Appendix A

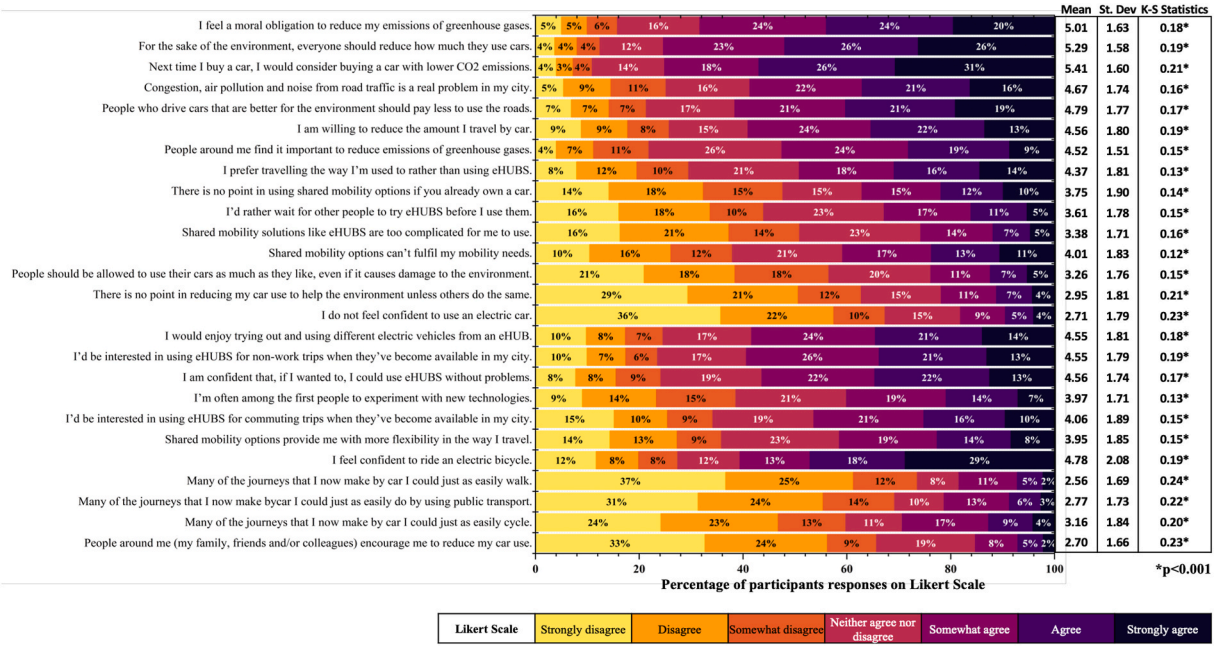


Fig. A.1. Participant responses on questionnaire survey

Appendix B

Table B.1
Discriminant validity: Fornell-Larcker criterion

Latent variable	Fornell-Larcker criterion				
	Alt. mode acceptance	Environmental Belief	Intention	Negative Attitude (S.M)	Positive Attitude (S.M)
Alt. mode acceptance	0.761				
Environmental Belief	0.337	0.730			
Intention	0.406	0.355	0.813		
Negative Attitude (S.M)	−0.133	−0.301	−0.434	0.726	
Positive Attitude (S.M)	0.360	0.536	0.653	−0.514	0.758

Table B.2
Discriminant validity: Cross loading

Observed variable	Cross-loading			
	Environmental Belief	Negative Attitude (S.M)	Positive Attitude (S.M)	Alt. mode acceptance
Att18	**0.803**	0.411	−0.229	0.187
Att19	**0.803**	0.426	−0.235	0.181
CarAtt1	**0.763**	0.470	−0.325	0.378
Att13	**0.623**	0.304	−0.149	0.223
Att17	**0.634**	0.306	−0.108	0.206
Att2	0.491	**0.851**	−0.439	0.260
Att7	0.447	**0.832**	−0.425	0.267
Att1	0.407	**0.801**	−0.396	0.302
Att12	0.329	**0.611**	−0.348	0.248
Att8	0.422	**0.762**	−0.411	0.351
Att5	0.314	**0.658**	−0.296	0.191
Att3	−0.288	−0.495	**0.854**	−0.184
Att9	−0.281	−0.420	**0.798**	−0.105
Att10	−0.188	−0.342	**0.708**	−0.129
Att4	−0.167	−0.332	**0.652**	0.023
Att6	−0.072	−0.156	**0.595**	0.051
CarAtt4	0.282	0.354	−0.159	**0.833**
CarAtt5	0.221	0.233	−0.057	**0.758**
CarAtt3	0.169	0.174	0.029	**0.739**
CarAtt7	0.315	0.287	−0.157	**0.710**

Table B.3
Discriminant validity (HTMT)

Latent variable	HTMT values				
	Alt. mode acceptance	Environmental Belief	Intention	Negative Attitude (S.M)	Positive Attitude (S.M)
Alt. mode acceptance					
Environmental Belief	0.402				
Intention	0.492	0.429			
Negative Attitude (S.M)	0.184	0.338	0.478		
Positive Attitude (S.M)	0.424	0.645	0.773	0.587	

Appendix C

Table C.1
Variance Inflation factor (Latent Variable)

Latent Variable	Alt. mode acceptance	Environmental Belief	Intention	Negative Attitude (S.M)	Positive Attitude (S.M)
Age	1.004	1.004	1.063	1.004	1.004
Alt. mode acceptance			1.195		
Environmental Belief			1.480		
Gender	1.004	1.004	1.053	1.004	1.004
Income	1.003	1.003	1.011	1.003	1.003
Intention					
Negative Attitude (S.M)			1.368		
Positive Attitude (S.M)			1.829		

Data availability

The data that has been used is confidential.

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